

Generative AI in Healthcare Applications: SLM and RAG

1. ABSTRACT

The exponential growth of medical knowledge and healthcare policies makes navigating domain-specific documentation challenging for both professionals and citizens. Large Language Models (LLMs) can answer queries, but generic models often suffer from hallucination and lack up-to-date, specialized knowledge. This project, **AI Sarthi**, introduces a robust Healthcare Assistant that combines a fine-tuned Small Language Model (SLM) with a Retrieval-Augmented Generation (RAG) pipeline. Utilizing microsoft/Phi-3-mini-4k-instruct as the base model, we employ PyTorch-based QLoRA (Quantized Low-Rank Adaptation) fine-tuning to adapt the model to healthcare instructions. To ensure responses are factual and verifiable, the system retrieves relevant context from a local ChromaDB vector store powered by BAAI/bge-small-en-v1.5 embeddings. A Streamlit chatbot serves as the user interface, generating accurate, context-aware answers accompanied by source citations. Evaluation frameworks measure latency, semantic similarity (BERTScore), and lexical overlap (ROUGE-L, BLEU), demonstrating the system's viability for resource-efficient, grounded healthcare question answering.

2. INTRODUCTION

The integration of Artificial Intelligence in healthcare offers transformative potential for democratizing medical information. However, traditional search systems often fail to synthesize complex policy documents, while standard conversational LLMs are prone to hallucinating facts—a critical risk in healthcare. To mitigate this, Retrieval-Augmented Generation (RAG) grounds the language model's generation in verifiable facts. Concurrently, Small Language Models (SLMs) have emerged as resource-efficient alternatives to massive LLMs, capable of running on consumer hardware while delivering high performance when fine-tuned. This project addresses the need for an accurate, locally deployable healthcare assistant by leveraging a fine-tuned Phi-3 Mini model and a dynamic RAG architecture to process and retrieve domain-specific healthcare documents.

3. PROBLEM STATEMENT

Navigating extensive Indian healthcare documentation and medical reports is time-consuming and complex. Generic AI chatbots lack the specific domain knowledge required to answer healthcare policy questions accurately and often generate plausible but incorrect information. There is a need for a resource-efficient, domain-adapted AI system that provides highly accurate, verifiable answers grounded in authorized healthcare documents without requiring massive computational infrastructure.

4. OBJECTIVES

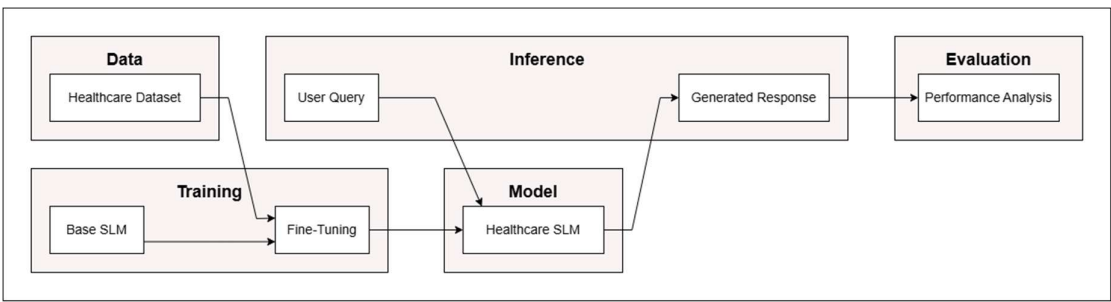
- **Healthcare Document Processing:** Ingest and preprocess healthcare PDFs and CSVs into clean, searchable chunks.
- **Retrieval-Based Grounding:** Implement a local vector database to perform semantic retrieval of medical and policy facts.
- **Domain-Specific Model Adaptation:** Fine-tune an SLM (Phi-3-mini-4k-instruct) using QLoRA for healthcare question answering.
- **Citation-Backed Responses:** Automatically extract and display source metadata alongside generated answers.
- **Interactive Chatbot:** Develop a professional Streamlit web interface with session history.
- **Evaluation:** Benchmark the model's generation quality, latency, and retrieval accuracy.

5. EXISTING SYSTEM

Existing healthcare information retrieval systems typically rely on keyword-matching (lexical search), which fails to understand the semantic intent of complex medical queries. On the other hand, relying purely on generic LLMs (such as ChatGPT without RAG) risks hallucinations and lacks real-time knowledge of specific regional healthcare policies. Furthermore, large models without domain adaptation require substantial cloud infrastructure, raising privacy and cost concerns.

6. PROPOSED SYSTEM

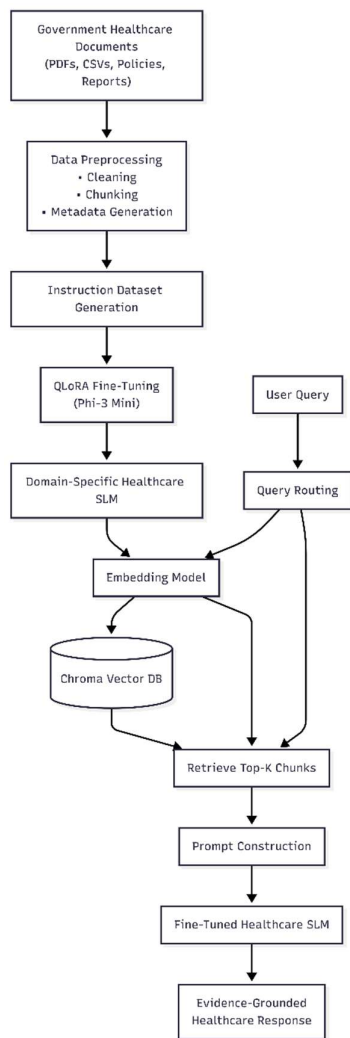
The proposed system, AI Sarthi, overcomes these limitations by combining a lightweight, fine-tuned SLM with a local RAG pipeline. The system preprocesses healthcare documents, embeds them using BAAI/bge-small-en-v1.5, and stores them in ChromaDB. Upon receiving a query, it retrieves the top-k most relevant chunks. A customized prompt combining the query and the retrieved context is fed into the QLoRA fine-tuned Phi-3 Mini model. The resulting answer is presented to the user through a Streamlit UI, complete with citations mapping back to the exact source documents.



7. LITERATURE REVIEW

- **Small Language Models (SLMs):** Models like Microsoft's Phi-3 series demonstrate that high-quality training data can yield capabilities rivaling much larger models while maintaining a tiny footprint.
- **Retrieval-Augmented Generation (RAG):** RAG architectures address LLM hallucinations by injecting retrieved, verifiable facts into the context window, bypassing the need to encode all knowledge directly into model weights.
- **PEFT and QLoRA:** Parameter-Efficient Fine-Tuning, specifically Quantized LoRA, enables the adaptation of language models on consumer GPUs by freezing the base model and updating only a small set of adapter weights.
- **Vector Databases:** Systems like ChromaDB facilitate high-speed, dense vector similarity searches, crucial for semantic information retrieval.

8. SYSTEM ARCHITECTURE



9. DATASET AND PREPROCESSING

- **Data Source:** Healthcare documents and policies (e.g., Health_Ministry_Annual_Report_25-26, NFHS-5_National_Report).
- **Processing:** Raw PDFs/CSVs are parsed and chunked.
- **Instruction Dataset:** A synthetic QA instruction dataset was generated and refined.
- **Dataset Statistics:**
 - Total samples: 13,047
 - Valid samples used: 3,979
 - Train/Validation/Test split: 3,822 / 82 / 75
 - Unique source documents: 8
- **Embeddings:** Text chunks are embedded using BAAI/bge-small-en-v1.5 and stored in a local ChromaDB collection (healthcare_chunks).

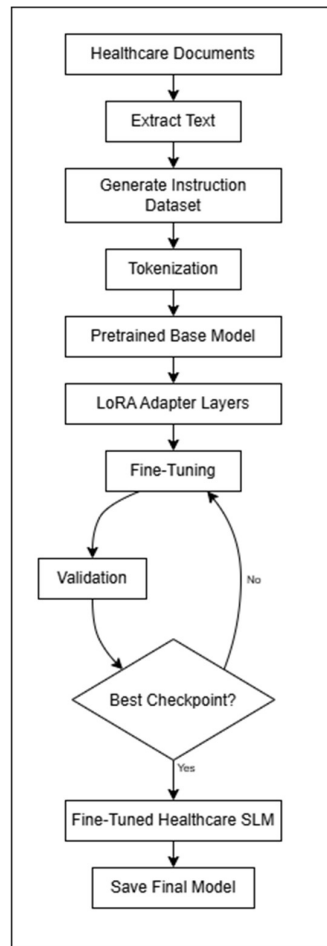
10. SLM SELECTION

microsoft/Phi-3-mini-4k-instruct was selected as the base model. Its parameter efficiency allows local deployment and cost-effective QLoRA fine-tuning. The 4k-instruct variant natively supports the multi-turn conversational format necessary for a chatbot, and its robust foundational training makes it highly receptive to healthcare-specific adaptation.

11. FINE-TUNING METHODOLOGY

The fine-tuning architecture utilizes PyTorch and PEFT (Parameter-Efficient Fine-Tuning).

- **Method:** QLoRA (Quantized Low-Rank Adaptation).
- **Base Model:** microsoft/Phi-3-mini-4k-instruct.
- **Dataset Preparation:** The cleaned synthetic QA pairs were formatted into the Phi-3 multi-turn chat template.
- **Process:** The script scripts/train_phi3.py orchestrates the training loop, optimizing the adapter weights for healthcare query comprehension and summarization.
- **Outputs:** The trained LoRA adapter weights are saved to models/outputs/.



12. RAG METHODOLOGY

- **Embedding:** BAAI/bge-small-en-v1.5.
- **Vector Store:** ChromaDB (vector_store/ directory).
- **Retrieval:** The system searches the vector store using the user's query and retrieves the top_k=15 (default) most semantically similar text chunks.
- **Generation:** A prompt is constructed containing the user query and the retrieved textual context. The fine-tuned SLM processes this prompt to produce a grounded response.
- **Citations:** Source metadata associated with the retrieved chunks is extracted and displayed to the user for verification.

13. IMPLEMENTATION

The project is modularized into distinct domains:

- **src/ingestion/**: Handles parsing PDFs/CSVs and text chunking.
- **src/vectorstore/**: Manages ChromaDB indexing and persistence.
- **src/model/training/**: Encapsulates the PyTorch QLoRA training loop and configuration.
- **src/rag/rag_pipeline.py**: Coordinates the retriever, prompt builder, and LLM inference.
- **src/evaluation/**: Contains metric calculation logic (BLEU, ROUGE, BERTScore).

14. STREAMLIT CHATBOT

The application features a professional, interactive UI built with Streamlit (`app/streamlit_app.py`).

- **Interface**: Maintains session state for chat history.
- **Integration**: Calls the `RAGPipeline` upon user input.
- **Features**: Displays the generated answer and lists the exact source files dynamically extracted from the vector store metadata.

15. EXPERIMENTAL SETUP

- **Hardware**: Consumer-grade hardware with GPU acceleration (CUDA) for model inference and training.
- **Software Stack**: Python, PyTorch, Streamlit, LangChain (for text splitting/vector operations), ChromaDB, Sentence Transformers.
- **Base Model**: microsoft/Phi-3-mini-4k-instruct.

16. EVALUATION

The evaluation framework (`scripts/evaluate_model.py`) measures system performance across multiple dimensions:

- **Latency**: Time taken to retrieve context and generate a response.
- **Lexical Overlap**: BLEU score and ROUGE-L f-measure via `nlk` and `rouge-score`.
- **Semantic Similarity**: BERTScore F1, assessing semantic closeness between generated and reference answers.
- **Manual Evaluation**: A template (`create_manual_evaluation_template.py`) measures Faithfulness (1-5) and Completeness (1-5).

17. RESULTS AND DISCUSSION

- **Training Results:** Training plots (loss.jpeg, learning_rate.jpeg, grad_norm.jpeg) in plots/training_plots/ confirm successful convergence of the LoRA adapter.
- **Dataset Quality:** Quality distribution analysis (plots/quality_distribution.png) indicates an average quality score of 4.38/5 for the generated instruction dataset.
- **Evaluation Metrics:** *Specific automated evaluation scores (BLEU, ROUGE, BERTScore) on the test set are not explicitly documented in the current implementation outputs.*
- **Discussion:** The integration of RAG significantly reduces hallucination risk by anchoring the SLM output directly to the retrieved top_k chunks.

18. LIMITATIONS

- **Vector Search Constraints:** Pure semantic search may struggle with highly specific keyword queries or numerical lookups without hybrid search (BM25 + Semantic).
- **SLM Capacity:** While highly capable, Phi-3 Mini may lack the deep reasoning capabilities of much larger parameter models (e.g., GPT-4) for highly complex medical logic.
- **Hardware Dependency:** Optimal latency during generation and local training requires dedicated VRAM (GPU).

19. FUTURE SCOPE

- **Hybrid Search Engine:** Integrating keyword-based BM25 search alongside ChromaDB for better precise term retrieval.
- **Advanced RAG:** Implementing query routing, re-ranking models (e.g., Cross-Encoders), and document hierarchical retrieval.
- **Broader Datasets:** Ingesting a wider array of regional Indian healthcare policies and multi-lingual translation support.

20. CONCLUSION

AI Sarthi successfully demonstrates the deployment of a highly specialized, localized AI Healthcare Assistant. By tightly integrating a PyTorch QLoRA fine-tuned Phi-3 Mini model with a ChromaDB-backed RAG pipeline, the system reliably answers complex domain queries. The architectural design guarantees that answers are grounded in factual sources, mitigating the risks of hallucination while maintaining the efficiency of an SLM, presenting a scalable solution for healthcare information retrieval.

21. REFERENCES

- *Microsoft. Phi-3 Technical Report: A Highly Capable Language Model Locally on Your Phone (2024).*
- *Dettmers, T., et al. QLoRA: Efficient Finetuning of Quantized LLMs (NeurIPS 2023).*
- *Lewis, P., et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (NeurIPS 2020).*
- *Hu, E. J., et al. LoRA: Low-Rank Adaptation of Large Language Models (ICLR 2022).*

22. APPENDIX

- Training configuration defaults can be found in `src/config/settings.py`.
- Dataset analysis scripts output to `plots/` and `reports/dataset_report.md`.